Winnow & Perceptron

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Winnow

- The winnow algorithm was developed in 1988, for when most of the features of a vector are not relevant.
- Assumption: The presence of even one relevant feature implies +.

	F_1	F_2	F_3	F_4	label
Person	old	male	tall	resident	Coffee
X_1	1	0	1	0	+
X_2	0	1	0	1	+
X_3	1	1	0	0	-
X_4	0	1	0	0	-

Winnow

- Notation: n is the number of total features, r is the number of important features. $F_j(X_i)$ is feature F_j of point X_i.
- Algorithm:
  - 1. set $w_1,\ldots,w_n = 1$. weight w_j corresponds to F_j
  - 2. For each vector X_i
    - If $\sum_{j=1}^{n} w_j F_j(X_i) \geq n$ guess + for coffee
    - Else guess – for coffee
    - 3. If we made a mistake on X_i
      - **A**. If we guessed + but – was true,

        for all j such that $F_j(X_i) = 1$, set $w_j = 0$
      - **B**. If we guessed – but + was true,

        for all j such that $F_j(X_i) = 1$, set $w_j = 2w_j$
  - 4. Iterate on all points until no mistake is found

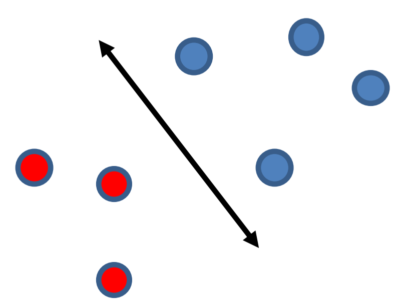
Winnow

- Claim: At most 2r log n mistakes
- Mistake B can happen at most r log n times.
  - After r log n mistakes, every important w_j has weight n.
- Mistake A can happen at most r log n times.
  - After mistake of type A, $\sum_j w_j$ decreased by at least n.
  - Mistake B adds to $\sum_j w_j$ less than n, total weight

    nr log n
  - # of mistakes of type A < r log n

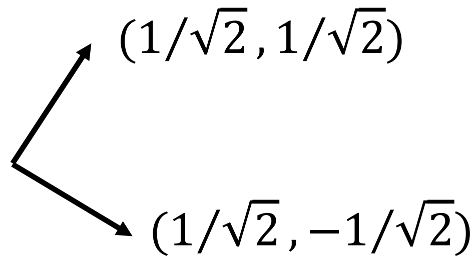
Perceptron algorithm

- We learned that hyperplanes with margin γ have VC-dimension $O(R^2/\gamma^2)$.
- How do you find a good hyperplane for set S?
- Brute force:
  - At most 3 points of S define the optimal hyperplane.
  - Try all combinations of 3 points out of n points of S
  - Check that no other point of S falls in the margin
  - $O(n^4) \rightarrow O(n^{d+2})$
- We'll see a better algorithm
  - Perceptron, 1958

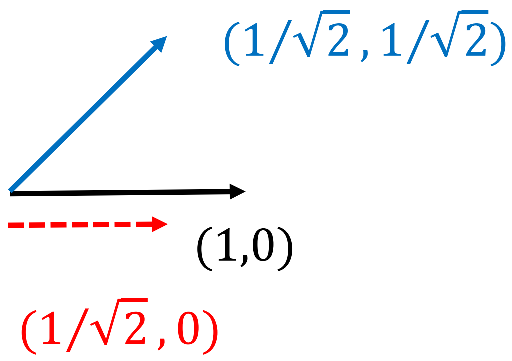


Dot product

- Properties of vector dot-product
  - Orthogonal vectors have product 0
  - Identical normalized vectors have dot product 1

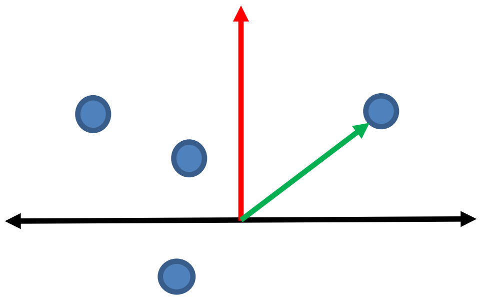


- Equal to cosign
  - Orthogonal projection



Dot product

- Denote hyperplane using normalized orthogonal vector
  - For any point, its distance to the line is its dot product with the vector



- Side note:
  - Take the k-dimensional basis vectors and their negation:
    - (1,0,0,…), (-1,0,0,…), (0,1,0,…), (0,-1,0,…),…,(0,0,…,1), (0,0,…,-1)
  - The best separating hyperplane is $\left(\frac{1}{\sqrt{k}}, \frac{1}{\sqrt{k}}, \frac{1}{\sqrt{k}}, \dots\right)$. This has margin $\frac{1}{\sqrt{k}}$

Perceptron

- Algorithm:
  - 1. Set vector $w_1=\vec{0}$ orthogonal to w* (not normalized)
  - 2. For round t=1,2,…
  - 3. Iterate over all points x_i
  - 4. If $w_t \cdot x_i > 0$ Guess +
  - 5. Else Guess –
  - 6. On mistake:
  - 7. If x is really +: $w_{t+1} \leftarrow w_t + x_i$
  - 8. If x is really -: $w_{t+1} \leftarrow w_t - x_i$
  - 9. exit round t
  - 10. If no mistakes this round, exit algorithm

Perceptron

- Claim: Perceptron finds a hyperplane consistent with S in $\frac{1}{\gamma^2}$ rounds,
  - Runtime $\frac{n}{\gamma^2}$ (where n is the number of points)
  - Assume points are in unit ball
- Let w* be the (normalized) hyperplane that separates the points of S with margin γ.
- Recall that w_t is not normalized…

Perceptron

- 1. We claim that

$$w_{t+1}w^* \geq w_t w^* + \gamma$$

- This means the vector gets closer to w*

  – If mistake x is +:

$$\begin{aligned} w_{t+1}w^* &= (w_t + x)w^* \\ &= w_t w^* + xw^* \\ &\geq w_t w^* + \gamma \end{aligned}$$

  – If mistake x is -:

$$\begin{aligned} w_{t+1}w^* &= (w_t - x)w^* \\ &= w_t w^* - xw^* \\ &\geq w_t w^* + \gamma \end{aligned}$$

Perceptron

- 2. We claim that $\|w_{t+1}\|^2 \leq \|w_t\|^2 + 1$
  - This means the vector norm doesn't get too large
  - If point x is +, and we guessed -:

$$\begin{aligned} \|w_{t+1}\|^2 &= \|w_t + x\|^2 \\ &= \|w_t\|^2 + 2w_t x + \|x\|^2 \\ &\leq \|w_t\|^2 + \boxed{2w_t x} + 1 \\ &< \|w_t\|^2 + 1 \end{aligned}$$

  - If point x is -, and we guessed +:

$$\begin{aligned} \|w_{t+1}\|^2 &= \|w_t - x\|^2 \\ &= \|w_t\|^2 - 2w_t x + \|x\|^2 \\ &\leq \|w_t\|^2 - \boxed{2w_t x} + 1 \\ &\leq \|w_t\|^2 + 1 \end{aligned}$$

Perceptron

- From claim 1, for any round M,

  $w_{M+1} w^* \geq M\gamma$

- From claim 2, for any round M,

  $\|w_{M+1}\|^2 \leq M \rightarrow \|w_{M+1}\| \leq \sqrt{M}$

- We have

  $(w_{M+1}/\|w_{M+1}\|)w^* \leq 1$

  $w_{M+1} w^* \leq \|w_{M+1}\|$

  $M\gamma \leq \sqrt{M}$

  $M \leq \frac{1}{\gamma^2}$

Perceptron

- The algorithm we saw only guarantees a separator.
- But we want a margin too!
  - Algorithm can be modified to get a margin arbitrarily close to γ with the number of mistakes still $O\left(\frac{1}{\gamma^2}\right)$.
  - Example for $\frac{\gamma}{2}$:

Perceptron

- Algorithm:
  - 1. Set vector $w_1 = \vec{0}$ orthogonal to w* (not normalized)
  - 2. For round t=1,2,…
  - 3. Iterate over all points x_i
  - 4. If $w_t \cdot x_i < \frac{\gamma}{2}$ but x is +
  - 5. $w_{t+1} \leftarrow w_t + x_i$
  - 6. exit round t
  - 7. If $w_t \cdot x_i > -\frac{\gamma}{2}$ but x is $-$
  - 8. $w_{t+1} \leftarrow w_t - x_i$
  - 9. exit round t
  - 10. If no mistakes this round, exit algorithm